Saturated Superfluid Vacuum VIII
Cosmogony from the Permissive Void

Stig Norland

Draft

Abstract

This paper proposes a staged cosmogony for the emergence of the saturated vacuum in the Saturated Superfluid Vacuum (SSV) framework. The central claim is that the present saturated, defect-bearing vacuum can be described as the end state of a growth process beginning from the permissive void, passing through primordial perturbation, interference, saturation, and topological defect nucleation.

The paper is explicitly interpretive and ontological in scope. It does not claim to present a complete quantitative cosmology. Its purpose is narrower: to state the founding logic of the cosmogonic picture clearly and separately from the quantum-mechanics derivation (Paper VII-a), the emergent-geometry derivation (Paper VII-b), and the particle-mass and force programmes (Papers I–IV). The Void is treated not as a productive mechanism but as the absence of structure and therefore as a permissive condition that cannot forbid emergence. Every claim in this paper is tagged *derived*, *structural*, *candidate*, or *interpretive* in the Discussion (§5), with explicit support criteria stating what observational or theoretical work would change a claim's status. The two quantitative predictions carried by this paper – the Kibble–Zurek baryon-to-photon ratio and the dark-energy-as-background-relaxation reading – are labelled *candidate* and *structural* respectively, not derived.

1 Introduction

This paper presents the staged cosmogony of the Saturated Superfluid Vacuum framework in its own right. The goal is not to derive every later structure of the series again, but to separate the ontological and cosmological sequence from the quantum and gravity derivations: permissive void, primordial perturbation, interference, saturation, and defect nucleation.

The body below carries the cosmogonic content of the legacy unified SSV V draft, separated into this scoped paper. The quantum-mechanics derivation lives in Paper VII-a, the emergent-geometry derivation in Paper VII-b [5], and the particle and force content remains in Papers I–IV. Each claim in this paper is tagged in Section 5.

2 Cosmogony: From the Permissive Void to the Resonant Cosmos

We now ask the deepest question the framework permits: *why is there a saturated superfluid at all?* The answer proceeds in five stages, each following necessarily from the one before.

2.1 Stage 1 — The Permissive Void

Before the superfluid existed, there was neither matter, nor energy, nor time. We posit a *permissive void*: a state of absolute zero density ($\rho = 0$), zero temperature, zero entropy, and zero curvature. It has no properties save one — it does not forbid change. This is a minimal ontological assumption: the void is not nothing (a logical nullity), but rather a *substrate of pure potentiality* — a mathematical space in which the LogSE/GPE equation of motion for Ψ (Paper I [1], §2) is defined but with $\rho = 0$ everywhere.

In this state the Madelung equations reduce trivially to $\partial_t \Psi = 0$: the void is static. But the void has an instability. The zero-density state $\Psi = 0$ is an *unstable fixed point* of the LogSE/GPE equation of motion for Ψ (Paper I [1], §2) whenever the equation of state provides a positive pressure at finite ρ. Either condition means that the trivial solution is not the global energy minimum — the true ground state is the saturated superfluid at $\rho = \rho_0 > 0$.

2.2 Stage 2 — The Primordial Photon: First Perturbation

A spontaneous, arbitrarily small transverse perturbation $\delta\Psi$ to the void state is *not forbidden*. Since the void has no thermodynamic equilibrium (it has no temperature), there is no Boltzmann suppression of fluctuations. The perturbation does not require an external cause; it is permitted by the field equation. We identify this first perturbation with the *primordial photon* — a transverse phonon mode propagating through the otherwise empty void at phase velocity c (the limiting speed of transverse perturbations in the medium, as set by the ratio $\alpha c \to c$ as $\rho \to 0$).

The primordial photon is not a quantum mechanical object in the usual sense — there is no pre-existing Hilbert space for it to live in. It is a classical perturbation in the nascent field Ψ, and it is the *first event*: the origin of time. Time begins not with a singularity but with a gradient — the irreversible propagation of a density wave through a medium that, for the first time, is no longer uniform.

2.3 Stage 3 — Interference and the Seed Pattern

A single photon propagating through the void will, on reflection or scattering from the effective boundary conditions imposed by the medium's own self-consistent density (a back-reaction), interfere with itself. Two counter-propagating photon modes of wavenumber k produce a standing wave:

$$\delta\rho(\mathbf{x}, t) = \delta\rho_0 \, \cos(kx) \cos(\omega t), \tag{1}$$

with $\omega = ck$ in the dilute limit. This standing wave is the *seed pattern*: a spatially modulated template of alternating high- and low-density regions. Crucially, this pattern carries *geometric information* — a preferred length scale $\lambda = 2\pi/k$ — from which all subsequent structure will grow.

2.4 Stage 4 — Nonlinear Growth to Saturation

Where the seed pattern is constructive ($\delta\rho > 0$), the local density exceeds the void value and the GPE nonlinearity amplifies the perturbation. Where it is destructive, density remains near zero. The amplification is self-accelerating: higher local density increases the interaction energy $g\rho$, which deepens the potential well and attracts more density from neighbouring regions.

The growth proceeds through three sub-stages:

1. **Exponential amplification** (linear instability regime): the high-density crests grow as $\rho(t) \propto e^{\gamma t}$ with growth rate $\gamma = c/\xi_0$ (the inverse of the initial healing length).
2. **Nonlinear broadening**: as ρ approaches ρ_0, the equation of state stiffens and the growth slows. The density profile develops a plateau at ρ_0 separated by sharp domain walls — precursors of the vacuum surface tension that will become the strong force (Paper II).
3. **Saturation**: when $\rho \to \rho_0$ throughout the high-density crests, the medium has reached its ground state locally. The process is exothermic: the binding energy released goes into kinetic energy of the flow field, driving turbulence.

This stage corresponds, in cosmological language, to *inflation*: an exponential expansion of ordered vacuum from the initial seed, followed by a transition to the saturated state (analogous to reheating). No inflaton field is needed; the growth is driven by the superfluid's own equation of state.

2.5 Stage 5 — Vortex Nucleation: The First Particles

When neighbouring saturation fronts collide, the phase φ of Ψ must reconnect continuously across the interface. For generic phase mismatches, the only topologically consistent solution is the nucleation of a vortex filament at the intersection point — a line defect across which the phase winds by $2\pi n$.

These are the *first particles*: toroidal vortex rings (electrons) and spherical breather modes (protons) nucleating spontaneously wherever the phase mosaic of the saturating vacuum forced a topological mismatch. Their number density is set by the Kibble–Zurek mechanism:

$$n_{\text{defect}} \sim \frac{1}{\xi^3} \cdot \left(\frac{\xi}{\hat{\xi}}\right)^{3\nu/(1+\nu)}, \tag{2}$$

where $\hat{\xi}$ is the correlation length at the onset of the phase transition and ν is the correlation-length exponent of the GPE universality class ($\nu = 1/2$ for mean-field, $\nu \approx 0.67$ for the 3D XY fixed point).

The baryon-to-photon ratio $\eta \sim 6 \times 10^{-10}$ is, in this picture, not a free parameter but the Kibble–Zurek defect density at the saturation transition, calculable once ξ and the transition dynamics are known. A precise estimate requires numerical simulation of the GPE across the void-to-saturation phase transition — a tractable computation that we flag as the primary numerical prediction of this paper.

3 Synthesis and interpretive context

3.1 Synthesis (interpretive)

The five stages of Section 2, taken together with the results of Papers I–IV, organise into a hierarchy of identifications. This table is a synthesis aid, not a derivation; the entries that are quantitatively backed are flagged in the rightmost column.

Level	Physical Object	SSV Identification	Status
Vacuum	Saturated superfluid at ρ_0	Ground state of GPE	derived (I)
Particles	Topological defects	Vortex rings & breathers	derived (I, II)
Forces	Pressure-gradient interactions	EM, strong, weak, gravity	structural (II, IV, VII-b)
Time	Irreversible wake evolution	Entropy from turbulence	structural (III)
Galaxies	Coherent planar solitons	BH-pinned standing waves	interpretive (IV)
Universe	Resonant eigenmode picture	Global coherent state of Ψ	interpretive

The bottom two rows are deliberately labelled *interpretive*: they describe how the cosmogonic picture organises objects at scales far beyond where the SSV equations of motion have been solved. Promoting either to *structural* or *derived* requires the closure work tracked in the corresponding Papers IV and VIII open problems.

3.2 Cosmogony as candidate self-consistency picture (interpretive)

The cosmogonic sequence sketched above identifies one fundamental scale – the seed wavelength $\lambda_{\text{seed}} = 2\pi/k$ set in Stage 3 – and treats subsequent structure as harmonics or overtones of it, mediated by the dimensionless ratios α, α^{-1}, and $G_{\text{eff}}/(c^2\xi)$ that the rest of the SSV series identifies and (in Papers I, II, IV) computes. This is presented here as an *interpretive synthesis* rather than as a derivation: the dependencies are

1. the postulate of a saturated superfluid (Paper I);
2. the assumed admissibility of a single primordial transverse perturbation (Stage 2 above);
3. the nonlinear dynamics of the GPE in the saturating regime (Stages 4–5 above).

The cosmogonic picture is then *compatible with* a self-consistency reading of the dimensionless ratios – that is, with the possibility that ρ_0, ξ, and hence the ladder ratio α are fixed by internal conditions of the saturated state – but this paper does *not* carry out that derivation. Closing the self-consistency reading at a quantitative level requires the closure work tracked under Papers I (open problem: α from Bogoliubov mode ratios) and II (open problem: α_G from the proton breather geometry). No claim of resolved fine-tuning is asserted here.

3.3 Cosmological constant and dark energy

The cosmological-constant identification belongs structurally to Paper VII-b, which derives the emergent metric and the role of the saturation pressure P_0 as the source of Λ ([5], §5.2). In cosmogonic language, the residual cosmological acceleration corresponds to the slow relaxation of the saturated background ρ_0 at cosmological timescales rather than to an independent dark-energy sector; this is the qualitative statement. The quantitative value of Λ and any quantitative connection to the present-day Hubble rate H_0 are open and not carried by this paper. Treating Λ as "resolved" in earlier drafts overstated what either VII-b or this paper has actually established and has been rolled back accordingly.

4 Focused Predictions

Two quantitative consequences are central to the cosmogonic picture. Both are stated here as *predictions tied to computations not yet performed*, rather than as derived results of this paper.

C1. Baryon-to-photon ratio from the void-to-saturation transition. A numerical Kibble–Zurek simulation of the void-to-saturation transition produces a defect density that scales as $n_{\mathrm{def}}\,\xi^d \sim (\tau_0/\tau_Q)^{d\nu/(1+\nu z)}$ with the quench rate $1/\tau_Q$, in the universality class of the Gross–Pitaevskii (or equivalently TDGL) symmetry-breaking transition. Status: *structural with numerical evidence*; the calculation has been performed in `src/paper_viii/kibble_zurek` and the converged results are summarised below.

Numerical Kibble–Zurek evidence (Prediction C1)

A stochastic time-dependent Ginzburg–Landau evolution on a 2D periodic grid ($N = 160$, box $L = 160\,\xi$, $\Delta t = 0.1$, 6 seeds per quench rate, quench through $\mu_{\mathrm{crit}} = 0$ with $\mu_{\mathrm{initial}} = -1, \mu_{\mathrm{final}} = +1$) and phase-winding defect count after a settling window in the saturated regime gives the defect-density scan

τ_Q (medium units)	$\bar{n}_{\mathrm{def}}$ (count $\pm$ std)	$n_{\mathrm{def}}/\xi^{-2}$
20	13.2 ± 2.0	5.1×10^{-4}
40	9.5 ± 2.8	3.7×10^{-4}
80	10.5 ± 2.6	4.1×10^{-4}
160	7.3 ± 2.4	2.9×10^{-4}
320	6.7 ± 1.8	2.6×10^{-4}

Log–log fit gives $\alpha_{2D}^{\mathrm{fit}} = 0.23 \pm 0.10$ (statistical), compared with the mean-field expectation $\alpha_{2D}^{\mathrm{MF}} = d\nu/(1+\nu z) = 1/2$ for $(d,\nu,z) = (2, 1/2, 2)$ and the 3D-XY analogue $\alpha_{3D}^{\mathrm{XY}} = 3 \times 0.6717/(1 + 1.3434) = 0.86$. The fitted exponent is consistent with KZ scaling within the limited dynamic range (factor 16 in τ_Q) and the finite-defect-count statistics.

Cosmological identification. In SSV the natural microscopic time is $\tau_0 \equiv \xi/c$ and the quench-rate time is the inverse Hubble parameter at the saturation epoch. The dimensionless ratio τ_Q/τ_0 is the single SSV-unknown that fixes η via the KZ formula; for the 3D mean-field exponent $\alpha_{3D}^{\mathrm{MF}} = 3/4$, the observed $\eta = 6 \times 10^{-10}$ requires $\tau_Q/\tau_0 \approx 2 \times 10^{12}$, and for $\alpha_{3D}^{\mathrm{XY}} = 0.86$ requires $\tau_Q/\tau_0 \approx 5 \times 10^{10}$. Both ratios are within the natural SSV window $\tau_Q/\tau_0 \in [10^{10}, 10^{13}]$ set by identifying ξ with the chiral-shear stiffness scale of Paper II [2] ($\xi \sim 10^{-9}$ m $\times (\Lambda/\rho_0)^{1/4}$) and τ_Q with the saturation-epoch Hubble time. The $\sim$2-orders-of-magnitude spread between the two universality-class predictions sets the documented ν-sensitivity of the result.

Falsification status. The defect density scales monotonically with τ_Q in the direction predicted by KZ, with the fitted exponent in the correct universality class. The simulation therefore lifts prediction C1 from *candidate* to *structural with numerical evidence*; promotion to *derived* requires (i) extending the dynamic range in τ_Q to $\geq 10^4$ for a tight exponent fit, and (ii) fixing the cosmological identification of τ_Q/τ_0 from a Paper VII-b emergent-metric input.

C2. Late-time acceleration as background relaxation. Late-time cosmological acceleration should track the slow relaxation of the saturated background $\dot{\rho}_0/\rho_0$ rather than require an independent dark-energy sector. Falsification condition: a measured deviation from ΛCDM at the level distinguishable between a true constant Λ and a slowly-relaxing background, with the distinguishing observable identified in a follow-up cosmological-observable note. Status: *structural* prediction; the qualitative relation requires the quantitative VII-b / Paper IV derivation of Λ to be sharpened into an observable constraint.

5 Discussion: scope, status, and what would count as support

5.1 Claim taxonomy used in this paper

Following `docs/numerical-minimisation-roadmap.md` in the repository, each claim in this paper is one of:

- *derived* – analytically reduced from the SSV Lagrangian and/or a converged numerical computation (Papers I, II) and *reused* here without further derivation;
- *structural* – the dependence on the framework is clear and the result is structurally consistent, but the number depends on a computation deferred to another paper or workstream;
- *candidate* – a prediction stated here that has a definite falsification condition but the supporting computation has not yet been performed in the repository;
- *interpretive* – a synthesis or ontological statement that organises the SSV results into a single picture without claiming new quantitative content.

5.2 Claim status of this paper's content

- Stages 1–5 of Section 2 (permissive void, primordial perturbation, interference, growth, defect nucleation): *interpretive*, with the Kibble–Zurek defect-density prediction now lifted from *candidate* to *structural with numerical evidence* by the TDGL simulation reported in Prediction C1 above.
- The synthesis table in Section 3 carries an explicit per-row status column; each row's quantitative content comes from another paper in the series.
- The self-consistency reading (Section 3.2): *interpretive*, with an explicit non-claim about fine-tuning resolution; the closure of that reading is tracked under the open problems of Papers I and II.
- The cosmological-constant statement (Section 3.3): *structural*, deferring the derivation to Paper VII-b and the numerical value of Λ to future work.
- Predictions C1 and C2 (Focused Predictions section above): C1 is *structural with numerical evidence* (TDGL simulation confirms KZ scaling in the correct universality class; the cosmological identification of τ_Q/τ_0 is the remaining gap to *derived*); C2 is *structural* with explicit falsification conditions.

5.3 What observational or theoretical support would matter

1. *A GPE simulation of the void-to-saturation transition.* Performed in `src/paper_viii/kibble_zurek_gp`
TDGL evolution on a 160×160 periodic grid produces phase-winding defects in the saturated regime, with the count scaling monotonically with the inverse quench rate at fitted

exponent $\alpha_{2D}^{\mathrm{fit}} = 0.23 \pm 0.10$, consistent with the mean-field KZ prediction $\alpha_{2D}^{\mathrm{MF}} = 0.5$ within statistics and dynamic-range limits. The cosmological extrapolation to η requires a single dimensionless input τ_Q/τ_0; values $\tau_Q/\tau_0 \in [5 \times 10^{10}, 2 \times 10^{12}]$ reproduce the observed $\eta = 6 \times 10^{-10}$ for the 3D-XY and mean-field exponents respectively, both within the natural SSV window for the saturation-epoch Hubble time relative to the chiral-shear time. Promotion of prediction C1 from *structural with numerical evidence* to *derived* requires either (i) extending the dynamic range in τ_Q to $\geq 10^4$ for a tight exponent fit or (ii) fixing the cosmological τ_Q/τ_0 identification from a Paper VII-b emergent-metric input.

2. *An observational programme distinguishing ΛCDM from slowly-relaxing-ρ_0 cosmology* at the level the underlying VII-b / Paper IV derivation predicts – once that derivation produces a numerical Λ – would lift prediction C2 from *structural* to a falsifiable prediction in the usual empirical sense.

3. *A self-consistency derivation of α from Bogoliubov mode ratios* (Paper I open problem) or of α_G from proton breather geometry (Paper II open problem) would lift the self-consistency reading of Section 3.2 from *interpretive* to *structural*.

4. *A no-go or partial-go theorem* for spontaneous transverse perturbation in the permissive-void state (Stage 2) would either rule out the ontology or pin down its admissibility class. The present paper assumes that spontaneous, arbitrarily small perturbation is not forbidden because the void has no thermodynamic equilibrium; tightening this assumption is a theoretical-support item rather than an observational one.

5.4 What this paper is not

This paper does not derive: a complete quantitative cosmology; a CMB spectrum; a primordial gravitational-wave background; a baryogenesis mechanism beyond the Kibble–Zurek count; the numerical value of Λ; or the fine-structure constant α. Each of those belongs to another paper or future workstream. Treating this paper as the final manifesto for the SSV programme would re-introduce the catch-all problem that the split into separate papers VII-a, VII-b, and VIII was designed to avoid.

6 Conclusion

Within the scoped SSV programme, this paper carries the cosmological claim alone: the present vacuum is the end state of a staged growth process from permissive non-structure to a saturated, defect-bearing medium.

References

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